

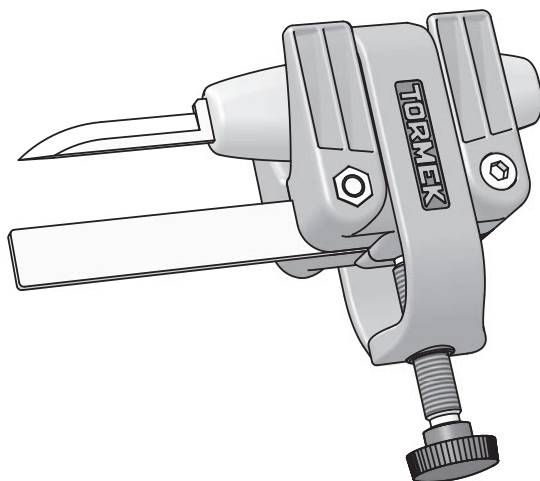
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SVM-00 Instruction



PATENT

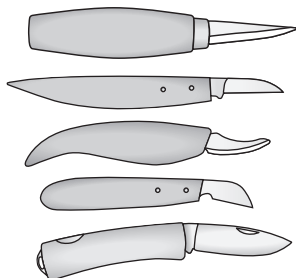
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SVM-00 Small Knife Holder

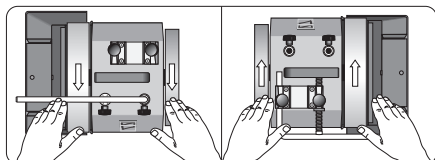


SMALL KNIVES

Various carving knives, skew knives and pocket knives.



Positioning of Machine



Sharpening direction: Towards or away from the edge.

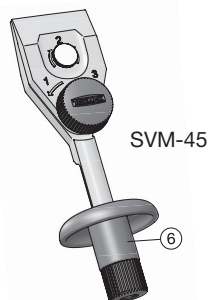
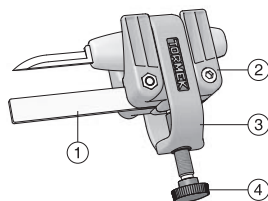
Note The height of the bench must not be too high when grinding towards the edge. 550–650 mm (22–26") is suitable depending on your height.

Design

The SVM-00 Small Knife Holder complements the SVM-45 Knife Jig. It extends the usage so that you can sharpen even the smallest knives.

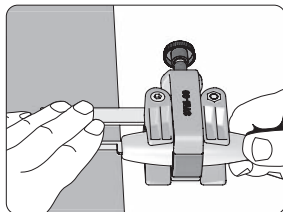
It consists of a Guide Bar (1), a Handle Grip (2), and an Adjustable Clamp (3). The knife handle is fastened with the Knob (4). The holder is designed to securely hold the knife without damaging the handle.

The Guide Bar (1) mounts in the jaws of the SVM-45 Knife Jig and you sharpen using the same technique as for the knife jig. The exact edge angle is set by turning the Adjustable Stop (6) or using the Micro Adjust on the Universal Support.

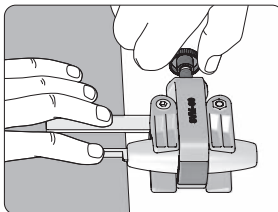


Mounting the knife

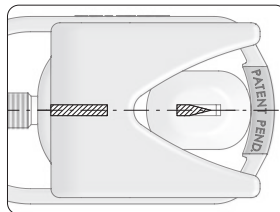
1. Mount the knife in the SVM-00 Small Knife Holder



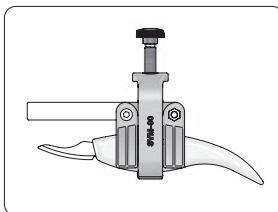
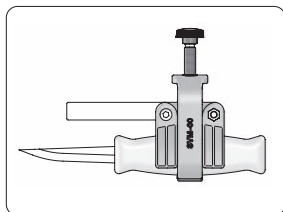
Mount the knife in the holder whilst lying flat on a table edge.



Press the Guide Bar and the knife blade against the table. Lock the knife with the Knob.

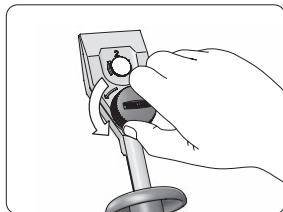


Check that the knife blade is in line with the Guide Bar.

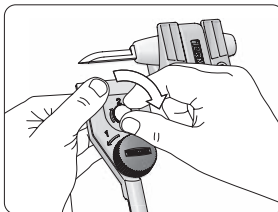


The shape of the knife handle determines how far into the holder the knife needs to be mounted. Make sure it is secured.

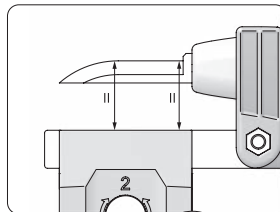
2. Mount the knife in the SVM-45 Knife Jig



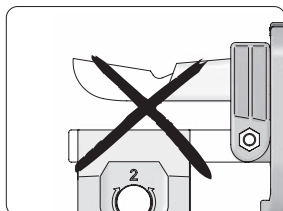
Check that the Knob is loose.



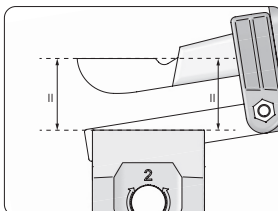
Mount the Guide Bar in the jig with the small adjustment screw.



Check that the knife edge is parallel to the SVM-45. Tighten the large black knob.



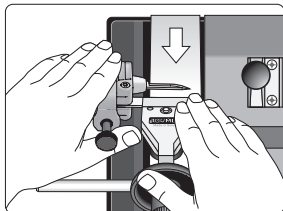
Wrong. The edge of the knife is not parallel to SVM-45.



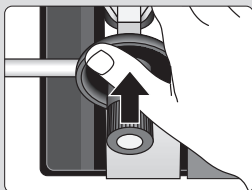
Right. The edge of the knife is parallel to SVM-45.

Sharpen away from or towards the edge?

Sharpen towards the edge



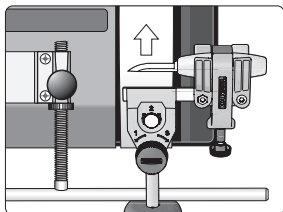
The Universal Support is normally mounted vertically for sharpening *towards* the edge. This will give you a more effective grind and you will get less burr to remove when honing. For folding knives (which cannot be locked) you should work in this position to prevent the knife blade from closing during the sharpening.



Important!

Apply a bit of pressure onto the stop so that it is always resting firmly against the Universal Support. In this way you will get an even edge and will not risk that the knife cuts into the grinding wheel.

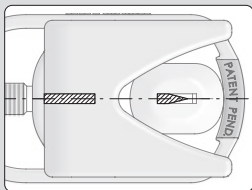
Sharpen away from the edge



You can also sharpen away from the edge with the Universal Support mounted horizontally. If your machine is placed on a bench over 700 mm high (27½") this will give a better working position. Some prefer this position since the knife cannot cut into the grinding wheel. It can also be advantageous if for some knives you want to have a slower grind.

Sharpening and honing

Set the edge angle and sharpen in the same way as the SVM-45 Knife Jig, see the Handbook. After sharpening, hone away the burr and polish the bevel on the honing wheel. The SVM-45 instruction is also available to download from www.tormek.com.



Note *Some knives are difficult to mount in line with the Guide Bar. Knives can also be sharpened with asymmetrical bevels. In these cases, and if the angle is not correct when you turn over to sharpen the second side, you can adjust it separately with the Micro Adjust.*

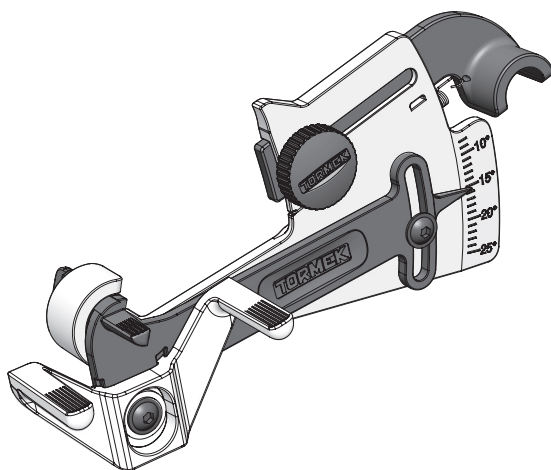
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KS-123 Knife Angle Setter Instruction

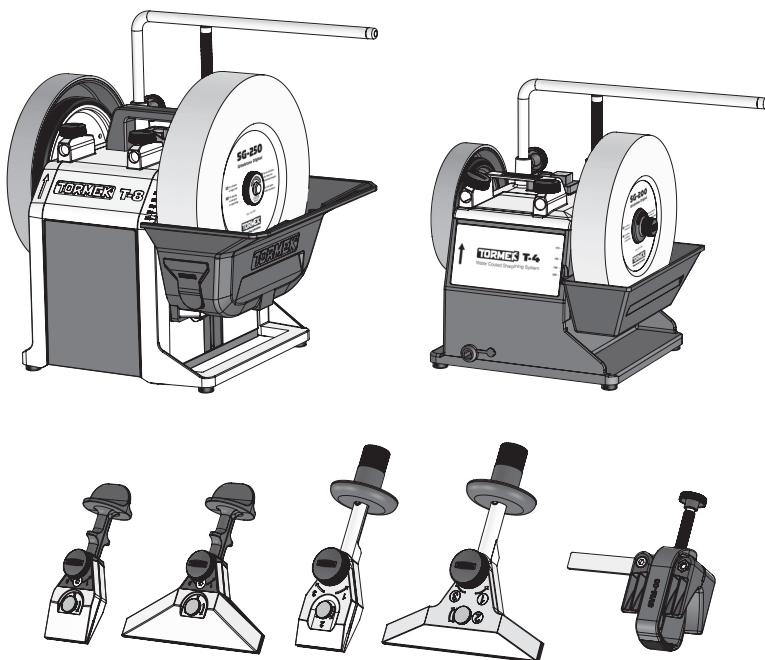


PATENT

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Compatibility

Compatible with the Tormek T-8 and Tormek T-4, as well as older machines with a universal support. Always used in combination with one of the Tormek knife jigs.

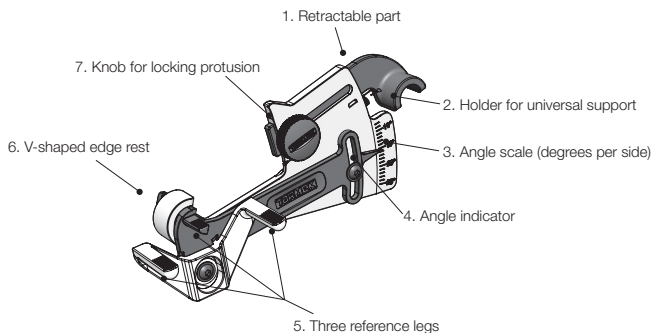


- KJ-45 Centering Knife Jig
- KJ-140 Wide Centering Knife Jig
- SVM-45 Knife Jig
- SVM-140 Long Knife Jig
- SVM-00 Small Knife Holder (together with KJ-45 Centering Knife Jig or SVM-45 Knife Jig)

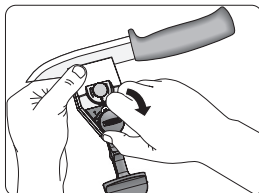
Instructional video



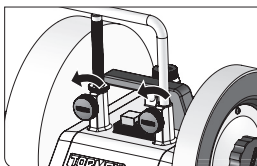
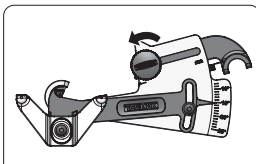
Tip! Scan the QR code to watch the instructional video on how to easily use the KS-123 Knife Angle Setter.



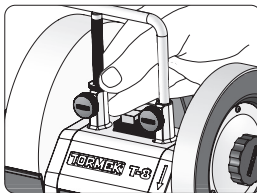
Preparations



1. Attach the knife to your knife jig according to its instructions and set them aside.



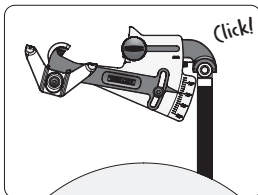
2. Make sure the knob for locking protrusion (7.) and the two knobs on the universal support mounts are loosened.



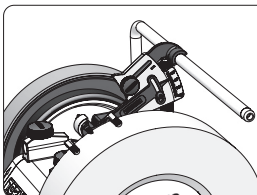
3. Lower the universal support slightly lower than you think you will need, as it will be easier to adjust the angle by raising the universal support. Leave the knobs on the universal support loosened.

Instructions

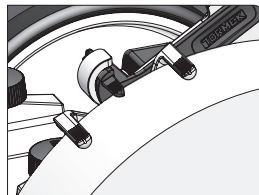
1. Mount the KS-123 Knife Angle Setter on your Tormek machine



Attach the KS-123 Knife Angle Setter to the universal support by snapping the holder for the universal support (2.) into place.

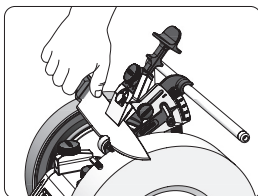


The three reference legs (5.) should rest over the grinding wheel, but don't have to all be in contact with the grinding wheel in this position.

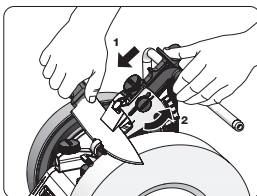


Note! The machine should be switched off during the entire setting process.

2. Adjust the protrusion of the angle setter to match your knife



Insert the knife blade with the edge **centered** in the V-shaped edge rest (6.) with the knife jig on the **right** side of the angle setter.



Slide the knife forward, along with the retractable part (1.) and set the stop of the knife jig against the universal support.

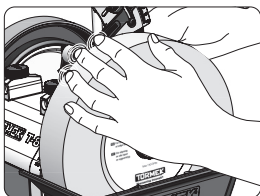
Make sure that the flat part of the knife jig stop rests against the support bar (like when you're sharpening), and tighten the knob for locking protrusion (7.) with the retractable part (1.) extended.

The KS-123 Knife Angle Setter is now adjusted to match your knife mounted in the knife jig. Put the knife jig with the knife to the side.

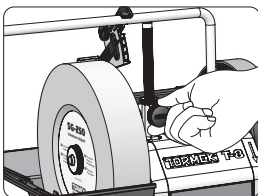
Note! Be careful not to slide the your knife against the grinding wheel.

Note! If you remove a lot of material from your knife you may need to redo this step before the final sharpening since the angle changes as the distance from the universal support to the apex changes.

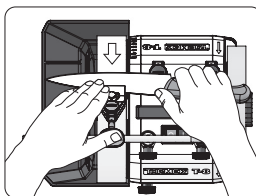
3. Setting the angle



1. Put one finger on each of the three reference legs (5.) with a light pressure. Raise the universal support until the angle indicator (4.) points to the desired sharpening angle on the angle scale (3.). Make sure that all three horizontal legs are resting against the grinding wheel in this step.

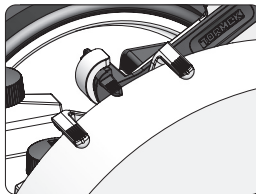
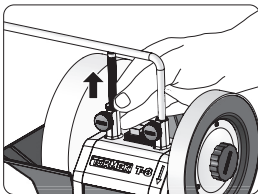
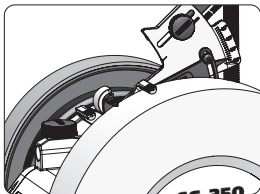


2. After raising the universal support so that the angle indicator points to the desired sharpening angle, lock the universal support knobs and remove the KS-123 Knife Angle Setter from the universal support.



3. The universal support is now set to the right height for the desired angle of the knife in the knife jig, and you can start sharpening your knife. The KS-123 Knife Angle Setter is self-adjusting for the grinding wheel diameter.

Note! If the universal support is initially set too low, the three reference legs (5.) will not be able to be in contact with the grinding wheel at the same time. Then you raise the universal support until the two outermost legs are resting against the grinding wheel, and start again with this step.



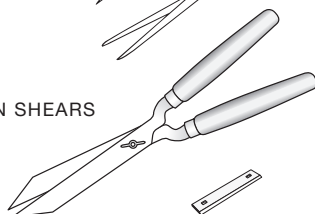
Scissors Jig SVX-150



SCISSORS

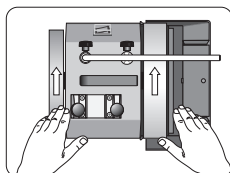


GARDEN SHEARS



PORTABLE ELECTRIC PLANER BLADES
(Not tungsten carbide)

Positioning of Machine



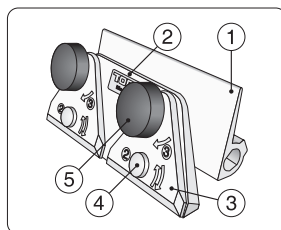
Grinding direction:
Towards the edge.

Tip The best working position is sitting on a chair in front of the machine.

Design

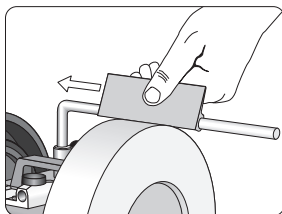
The jig comprises a support plate (1) and a holder (2) with two clamps (3) for the scissors. Usually you use only the right clamp. For large scissors and garden shears you use both clamps.

You adjust the clamp to fit the scissors' thickness with the screw (4). You lock the scissors by tightening the knob (5). The support plate has a sliding surface, so the holder with the scissors can easily be moved while it is pressed onto the plate.

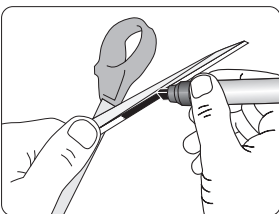


Setting the edge angle

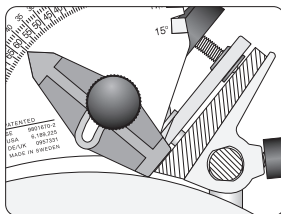
The edge angle is set by adjusting the support plate. You can either repeat the existing angle or put a new edge angle to your scissors by using the AngleMaster WM-200. A common edge angle is 60°.



Slide the support plate onto the Universal Support. Lock the Universal Support in the lowest possible position with clearance for the plate.

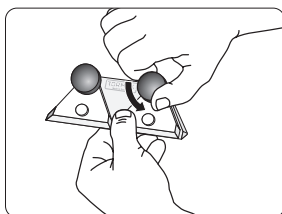


If you want to repeat the existing edge angle, colour the bevel with a marker to see where the grinding will take place.

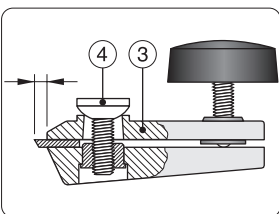


If you want to grind a new edge angle, use the AngleMaster WM-200. Let the indicator touch the holder between the two clamps.

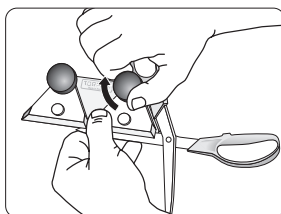
Mounting the scissors in the jig



Loosen the larger knob (5).

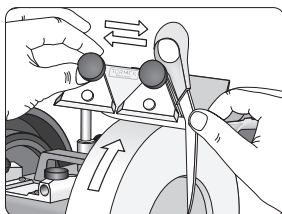


Adjust the screw (4) so the clamp (3) fits the scissor thickness. Leave about 3 mm ($\frac{1}{8}$ ") protruding.

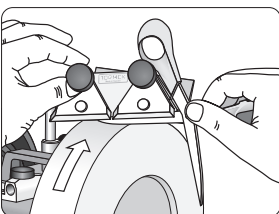


Ensure that the other blade will remain clear of the side of the wheel and tighten the large knob to hold the scissors in place.

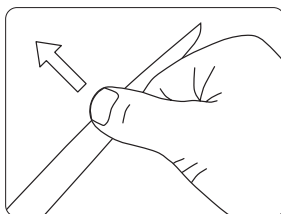
Grinding



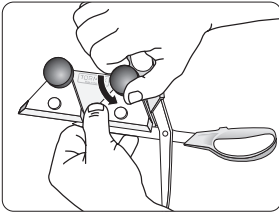
Grind carefully by placing the jig on the support plate with a light downward pressure.



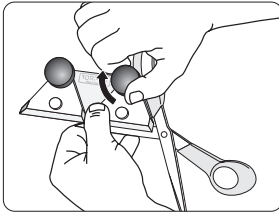
Move the jig back and forth across the wheel while following the shape of the blade.



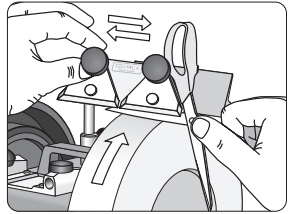
Grind until you have a burr along the entire length of the blade, and you can see an evenly ground bevel.



Remove the partially ground scissors from the jig by loosening the large knob.

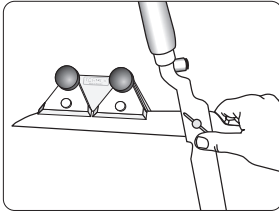
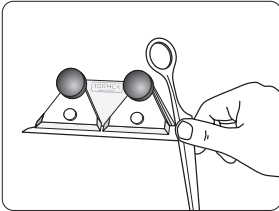


Mount the other blade in the jig.



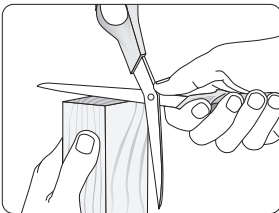
Grind the other blade in the same way as the first one.

Larger scissors and garden shears



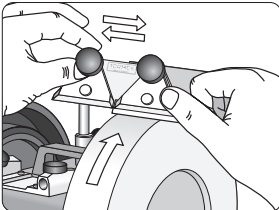
For larger scissors and shears, use both clamps.

No honing



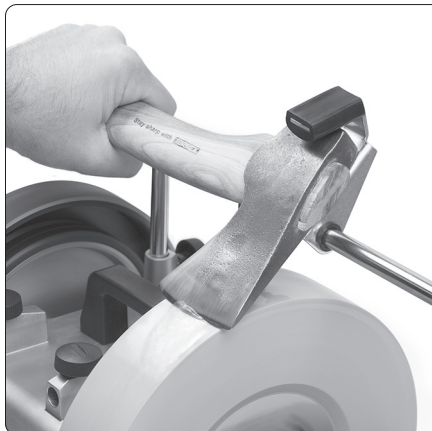
Do not hone the edge after grinding! There is no need for a finer surface. As a matter of fact a coarser surface on the scissors makes them work better, as the material (cloth or paper) to be cut slides less and cuts more easily. Remove the little burr resulting from the grinding by drawing the scissors through the end grain of a piece of wood.

Portable electric planer blades



These are ground the same way as scissors. The bevel and back should however be honed. After grinding, keep the blade in the holder and hone the bevel and the back alternately until the burr disappears.

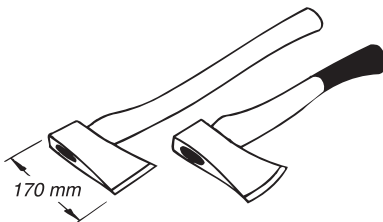
Axe Jig SVA-170



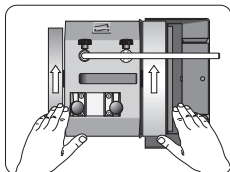
AXES AND HATCHETS

Fits axes up to 170 mm (6¾").

Larger axes are ground without a jig and supported on the Universal Support. (page 42).



Positioning of Machine

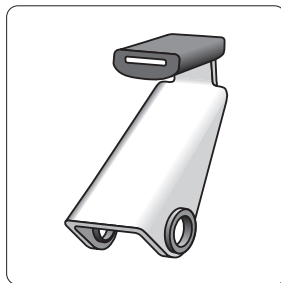


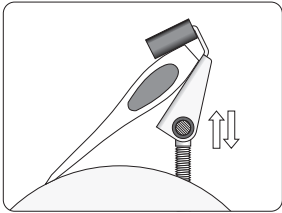
*Grinding direction:
Towards the edge.*

Design

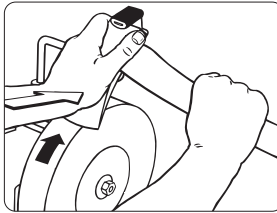
The jig comprises a clamp with a rubber lined, wedge shaped stop, which holds the axe in a fixed position towards the grindstone. The clamp works on the Universal Support, on which it slides across the wheel.

The jig is designed so that the rotation of the wheel presses the axe firmly into the jig. You only need to hold and guide the axe with your hands. This jig only works with the wheel running towards the edge. If necessary, the honing wheel can be removed to allow clearance for the axe handle.

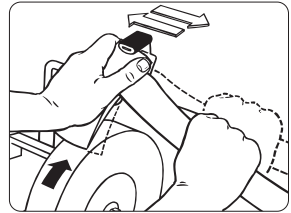




Place the axe into the jig. Set the edge angle by adjusting the Universal Support. Either at the original angle using the Marker Method or at a new angle using the AngleMaster WM-200.



Press with your palm on the axe, close to the wheel. The axe will automatically be pressed up against the stop in the jig and the grinding is most efficient.

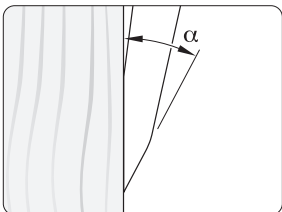


Follow the shape of the edge (straight or curved) during grinding by raising or lowering the handle. Let the jig slide sideways on the Universal Support so that the wheel wears evenly.

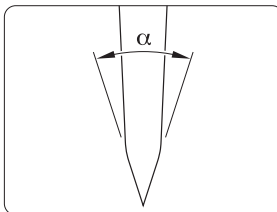
Edge Angle

The edge angle of a carpenter's axe or hatchet should be 25–30°, depending on the hardness of the wood. Grind the edge with a smaller bevel angle on the side towards the wood than on the other side. This longer bevel gives you a good support towards the wood. Also, as the axe works closer to the surface of the wood, the cutting will be more effective and easier to control.

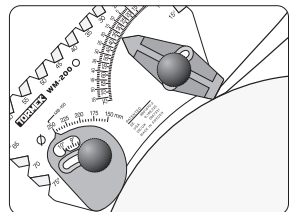
A felling axe and a hunting axe must have a larger edge angle, 30–40°, so that the edge is more durable. Round off the heels of the bevels and the axe cuts more easily. This is done by lowering the Universal Support and grinding in two steps.



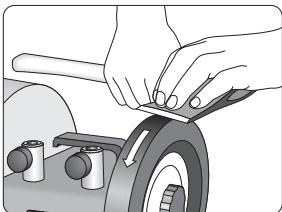
Put a longer bevel on the wood-side on a carpenter axe.



A felling axe should have an edge angle (α) of 30–40°. Round off the heels and the axe cuts more easily.



You can set and measure the edge angle with the AngleMaster WM-200.



Honing and polishing the bevels on the leather honing wheel makes the edge cut more easily and the sharpness will stay longer. Always hone away from the edge.

Short Tool Jig SVS-38



CARVING TOOLS WITH A STRAIGHT SHANK

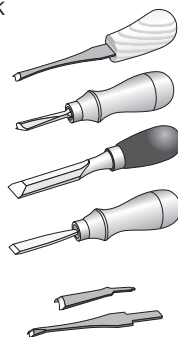
Straight gouges

Straight V-tools

Short wood chisels

Square carving chisels

Gouges and V-tools for reciprocating power carvers



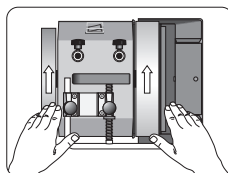
SVS-38 is a further development of SVS-32. It can handle slightly wider tools but apart from that, the function is the same as for SVS-32.

SVS-38: Max tool width 38 mm (1½").

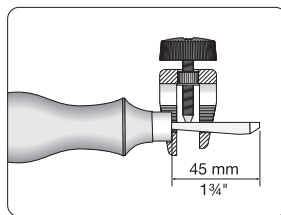
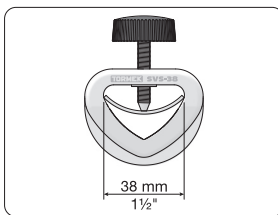
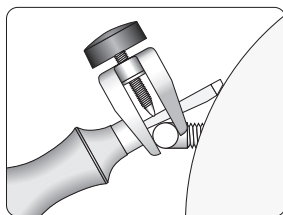
SVS-32: Max tool width 32 mm (1¼").

Min length of tool 45 mm (1¾") at 20° edge angle.

Positioning of Machine



*Grinding direction:
Away from the edge.*



Design

This patented jig was developed for short woodcarving tools, butt chisels and tools for power carvers. The jig has two parallel flanges running on both sides of the Universal Support.

The technique achieved with this design ensures that at all times the jig – even when rotated – holds the tool square to the wheel. This is a great advantage, as you can continuously concentrate on how the edge touches the grindstone or the honing wheel without having to worry whether the jig is square to the stone.

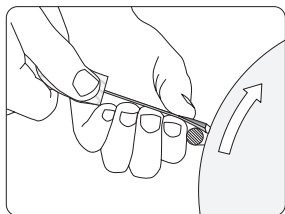
Carving Gouges

Grinding/Sharpening/Honing or Just Honing?

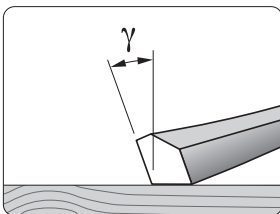
As mentioned in the chapter *Sharpening Techniques for Carving Gouges and V-tools* on page 20, you should question whether you should grind the tool or you should go directly to the honing wheel. This question is especially valid when coping with small and delicate tools, for which this jig is designed. The small size of the tool, which can be as small as 3 mm ($\frac{1}{8}$ "), also enables the honing wheel to remove small amounts of steel and thus substitute the grinding.

The following instruction describes the whole process for achieving a sharp edge on a tool, which has become so blunt that it needs re-grinding. The instruction is also valid for tools, upon which you want to put a new shape and for tools upon which you want to change the edge angle. A tool which is only slightly dull and which has the right shape and edge angle, you should not grind, but go directly to the honing process (page 71).

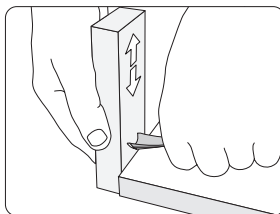
Shaping the edge



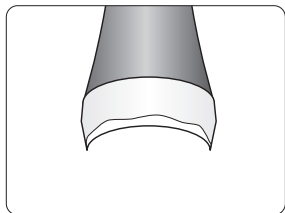
Shape the edge by resting the tool on the Universal Support placed horizontally and close to the wheel.



The edge plane angle (γ) should be approx. 20°. (Page 21).



Flatten and smooth the blunt edge with the fine side of the Tormek Stone Grader, SP-650.

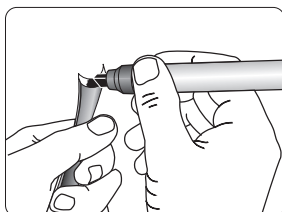


Now you have a line of light, which is your guide as to where to grind.

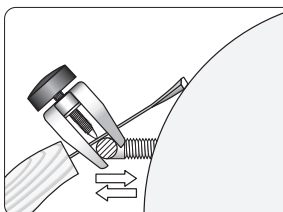
Edge Angle

As described in the chapter *Sharpening Techniques for Carving Gouges and V-tools* on page 24, the choice of edge angle is very important for a carving tool. The way of setting the edge angle depends on if you want to replicate an existing edge angle or if you want to put a new edge angle on your tool.

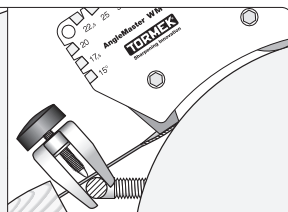
When replicating an existing edge angle, you should use the *Marker Method* (page 41). If you want to put a new edge angle to your tool, you can either set the angle by eye or you can use the AngleMaster (page 142).



When replicating the existing edge angle you should use the *Marker Method*.

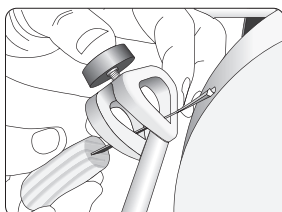


Setting a new edge angle can be done by eye or you can set to a pre-determined edge angle using the AngleMaster (page 142).

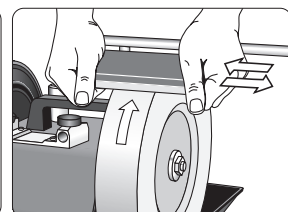
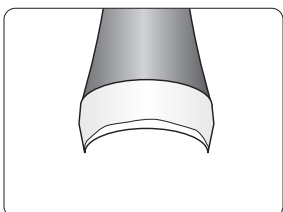


Grinding

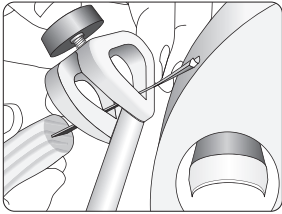
When you have put the right shape on the tool and set the edge angle, it is time for grinding. Use the *line of light method* described on page 20.



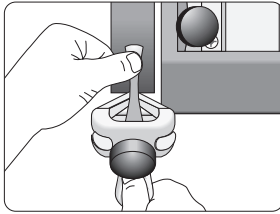
Always grind where the line of light is the thickest, while rolling the tool on the Universal Support. Check frequently where the grinding takes place. Grind until you get an even and thin line of light.



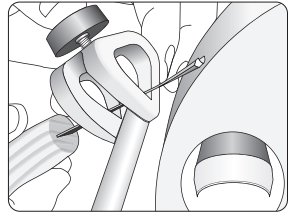
Grade the grindstone for fine grinding with the fine side of the Stone Grader SP-650.



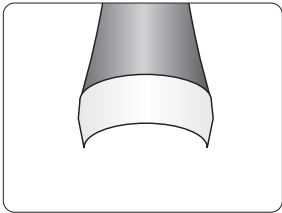
Continue grinding. Check the result frequently so you do not over grind on a spot.



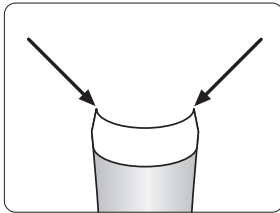
Remove the burr free-hand on the leather honing wheel to observe the line of light more clearly. The tool is left mounted in the jig.



Grind again. Now with a very light pressure. Check frequently so that you do not over grind on one spot.



Stop grinding immediately when the line of light disappears! This is the sign that the edge is sharp.

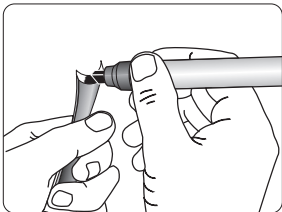


Be careful when you grind at the sides of the gouge, so you do not round off the corners.

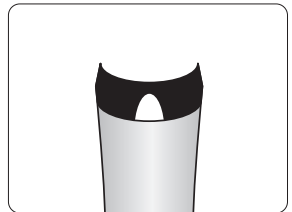
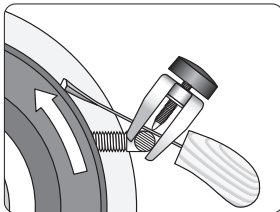
Important It is very easy to be misled by the burr and mistake the burr for the line of light! Therefore you must remove the burr frequently during the finishing of the grinding operation, so you clearly can watch the progress of gradually thinning the line of light.

Honing

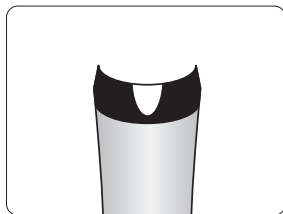
It is important that you hone with the same edge angle as used during the grinding. This is done by setting the Universal Support carefully using the *Marker Method*.



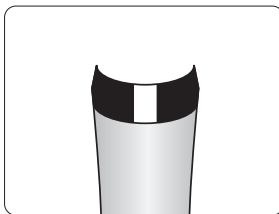
Colour the bevel with a marker and mount the tool in the jig. Set the Universal Support so that the honing wheel touches the entire length of the bevel. Check with the honing wheel running.



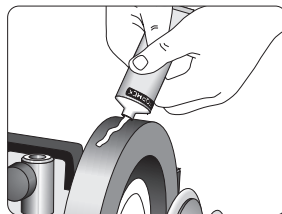
Wrong setting. The wheel is hitting the heel. Adjust the Universal Support away from the wheel.



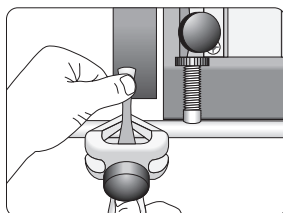
Wrong setting. The wheel is hitting the tip. Adjust the Universal Support towards the wheel.



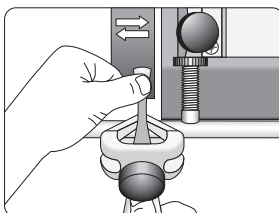
Correct setting. The honing wheel hits the full length of the bevel.



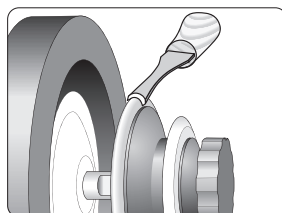
Prepare both the honing wheels with honing compound (p 44).



Sharpen/hone the edge. Roll the tool on the Universal Support and apply pressure on the tool near the edge.

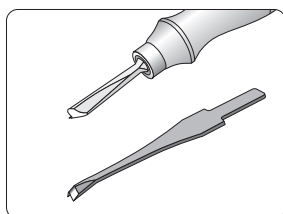


Move the jig sideways so you work on the whole width and avoid uneven wear on the leather.



Remove the tool from the jig and hone the inside free hand on the leather honing wheel.

V-tools



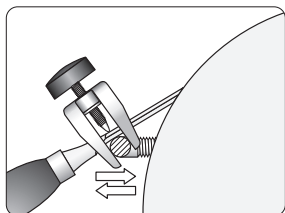
In the chapter of SVD-186 R on page 90 it is described how to grind and hone V-tools. This principle is the same when using the SVS-38 jig.

Short Wood Chisels and Carving Chisels

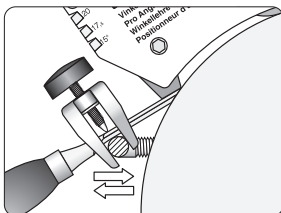
Grinding a wood chisel is much easier than grinding a gouge or V-tool, since a wood chisel has a single and straight edge. Before grinding, check that the edge is square across the edge. If not, make the edge square on the grindstone resting the tool on the Universal Support positioned close to the grinding wheel.

When grinding a new wood chisel for the first time, check that the back is flat and is free from manufacturing grooves. If necessary, flatten and smoothen it on the outside of the wheel (page 123) and then hone it on the leather honing wheel.

Setting the edge angle



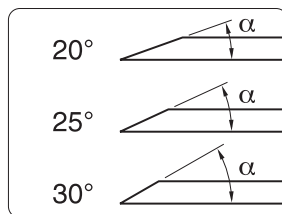
Setting the existing edge angle. Adjust the Universal Support so the entire bevel touches the wheel. Use the Marker Method.



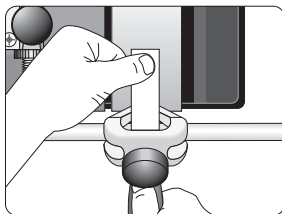
Setting a new edge angle. This can be done by eye or you can, as shown here, set the jig to a pre-determined edge angle using the AngleMaster WM-200.

Edge Angle

Wood chisels are usually ground with a 25° edge angle (α). If you work delicate details with a wood chisel in soft wood, you can decrease the edge angle down to 20°. If you work in hard wood and when using a mallet, you must increase the edge angle to 30° or even larger.

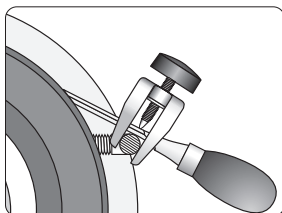


Grinding

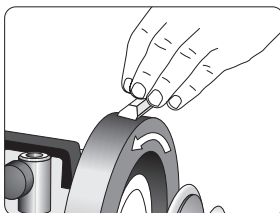
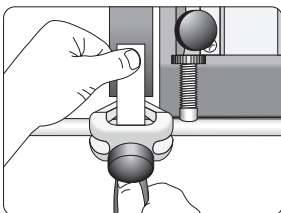


Press with your thumb close to the edge for best control. You also control with your hand on the handle that the edge is presented correctly to the grinding wheel and not turned. Check frequently so that you achieve a symmetric edge. Do not slide the jig sideways, but instead lift it when you move it to another spot on the wheel.

Honing



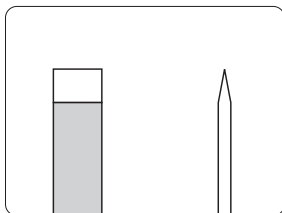
Keep the tool in the jig and move to the leather honing wheel. Set the Universal Support so that the honing angle is the same as the grinding angle. Use the Marker Method.hone away the burr and polish the bevels to a mirror finish.



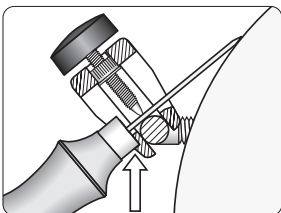
Hone the back free hand. Watch that you do not round the tip. Hold the tool so it is a tangent to the wheel.

Square Carving Chisels

These have symmetrical bevels on both sides. This jig copes with tools down to a blade length of approx 60 mm (2 $\frac{3}{8}$ ") at 25° edge angle. As the jig cannot be turned upside down (as the SVS-50 jig), you need to de-mount the tool, turn it 180° and re-mount it in the jig. Let the tool handle touch the jig during each mounting and you will get the same setting and symmetrical bevels.

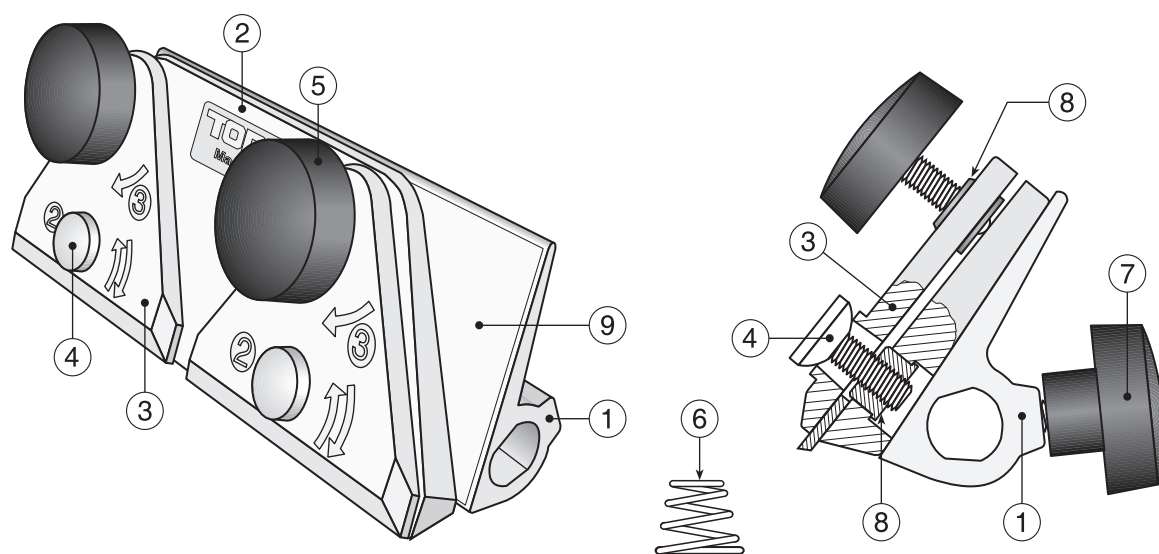


A square carving chisel.



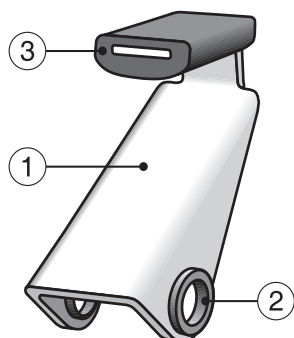
Let the tool handle touch the jig at both the mountings.

SVX-150 Scissors Jig



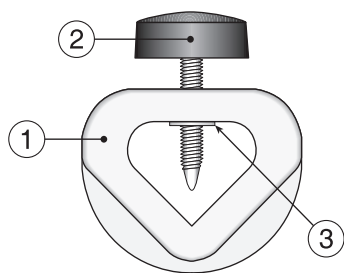
Item no.	Article no.	Denomination	Quantity	Weight/piece	
				g	oz
1	0230	Support plate	1	156	5,50
2	2030	Holder	1	126	4,44
3	2040	Clamp	2	31	1,09
4	1060	Setting screw	2	7	0,25
5	7050	Tightening screw	2	9	0,32
6	0500	Spring	4	0	0,00
7	7031	Locking screw	1	10	0,35
8	1050	Brass bushing, M6	4	4	0,14
9	9492	Sliding film	1	4	0,14

SVA-170 Axe Jig



Item no.	Article no.	Denomination	Quantity	Weight/piece	
				g	oz
1		Clamp	1		
2	3030	Rubber stop	1	15	0,53
3	3120	Bushing	2	1	0,04

SVS-38 Short Tool Jig



Item no.	Article no.	Denomination	Quantity	Weight/piece	
				g	oz
1		Body	1		
2	7070	Knob screw	1	13	0,46
3	1050	Brass bushing, M6	1	4	0,14